Undergraduate Research Project Report

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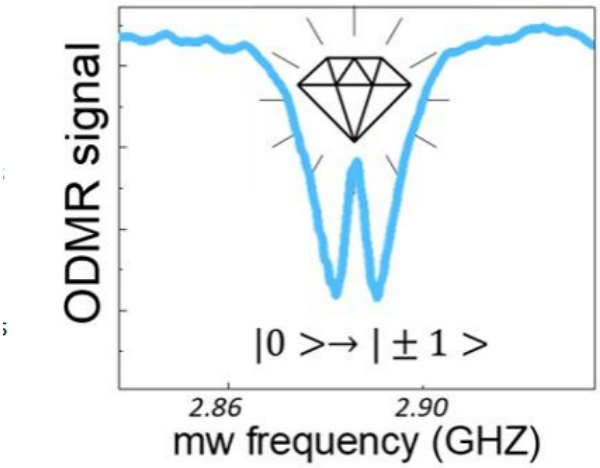
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1. Introduction

Nitrogen Vacancy centres are one of the many color centres present in Diamond. Their long spin coherence times at room temperature made them a promising candidate for high sensitivity magnetometry at room temperature.
Many schemes have been developed for this purpose [1], with of most them having few things in common: a diamond sample rich with NV centres, a green light source (532 nm) , a fluorescence collection setup for red light (637 nm) , a Microwave field generator and a static bias magnetic field.

When the input green light is allowed to fall on the NV centres for some time, the electrons are polarized to the ground state. The microwave field causes population inversion, and using the ODMR technique, the fluorescence vs frequency plot is obtained ,using which the magnetic field is determined.



The scope of my research is to develop an efficient scheme to excite the diamond sample with green light. This serves two purposes : the optical efficiency from input to output is improved , and the number of NV centres excited, which is an important parameter affecting sensitivity, can be precisely controlled with a good design.

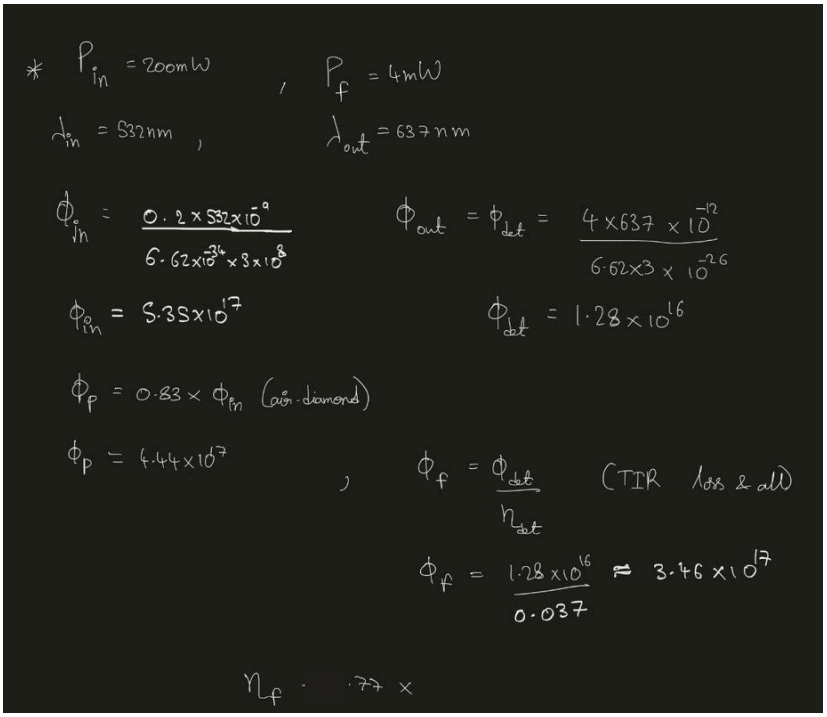
This report starts with the description of our current setup and follows with an analysis of where integrated optics can improve its performance. It follows with a discussion of the first tried method, an Extreme Mode Converter. The working principle, design choices and challenges with this method are discussed. Then, the next method of using an Ultra Efficient Grating Coupler is discussed and the simulation results are shown. The report concludes with discussion of further work which can be done following this.

2. Spatial Optics

a. Current setup

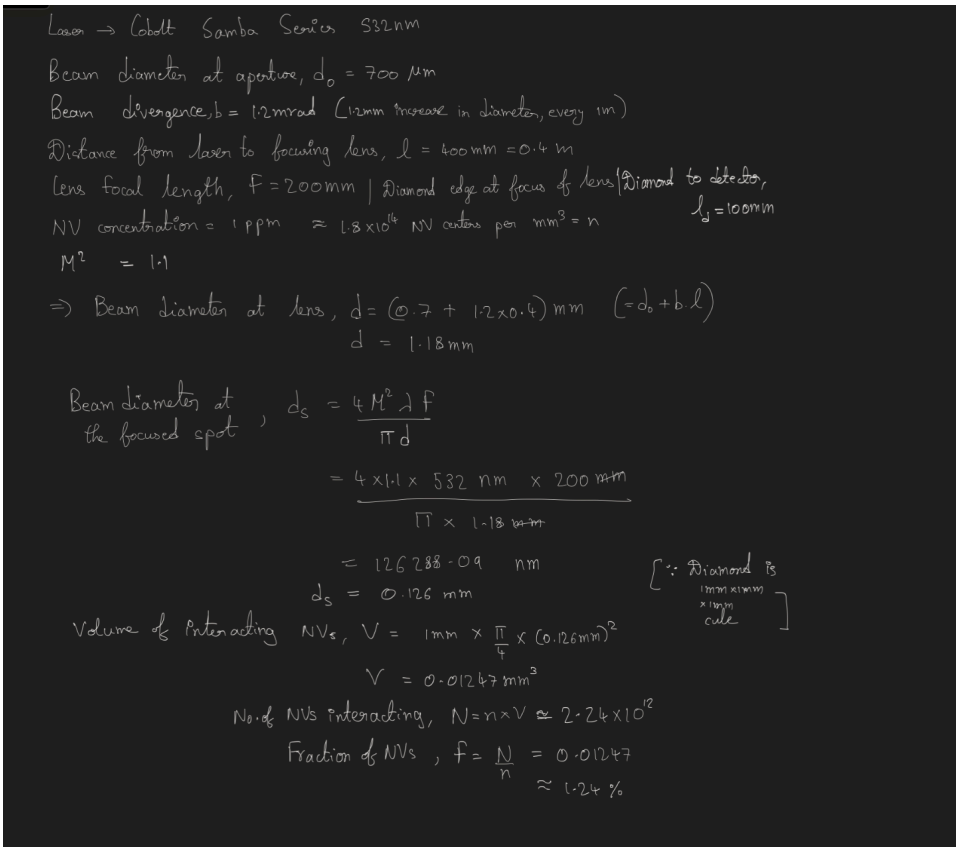
In this section, I will explain the current setup at IIT Madras. Through some simple calculations, some values are evaluated and the scope for improvement is identified.

Firstly, the number of red photons emitted by the diamond sample per green photon of input. Using the values of input power and detected fluorescence power, I evaluated the photon flux (no. of photons per sec) leaving the laser and the photon flux received at the detector. Further, I accounted for the fresnel reflection loss and TIR losses at the diamond air interface to calculate the green photon flux interacting with the NVs and the red photon flux emitted by the NVs as shown below



The η_f is found to be 0.77, which is a good value. In the case that this value is low, it can be optimized by structures which create long interacting path lengths in diamond, as explained in the paper on Light Trapping Diamond Waveguides [2].

Next, the fraction of NV centers interacting with the input light. This number is important as it directly affects the sensitivity of the magnetometer [3].
The beam diameter at the aperture is considered, and the expansion of beam is accounted for by the beam divergence value of the source.
Our setup uses a focusing lens with the diamond entrance at the focal point of the lens. Using the formula for the focused spot size, the diameter at the entrance to the diamond is obtained. Then, considering the dimensions of the sample, and the concentration of NV centers, the number of interacting NVs and the fraction are found out.



It is seen that the fraction of interacting NVs is very low. This is an important scope for improvement.

b. Benefits of Integrated Optics

Three are key problems that integrated optics can solve:

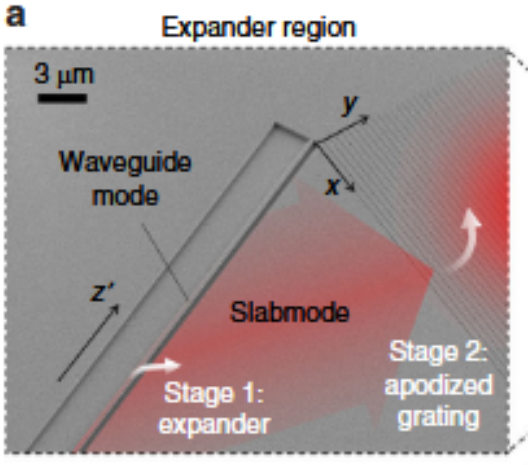
1. The current setup is bulky. For maximum utility of the magnetometer, a portable setup is needed, which can be realised by integrated optics
2. Low efficiency of light conversion. Due to fresnel reflection and TIR losses, a lot of the input light is getting wasted (wastage of power). This can be avoided by carefully designed integrated optical structures.
3. Most importantly, the volume of interaction cannot be easily controlled. This directly affects the performance of the magnetometer and can be fixed using an integrated optics approach.

In the next sections, some novel integrated optical structures are explored.

3. Design for the Extreme Mode Converter

a. Working Principle

Kim *et al.* [4] came up with a novel structure designed to convert a photonic waveguide mode to a free space Gaussian Beam, demonstrating an increase in model area > 10^5



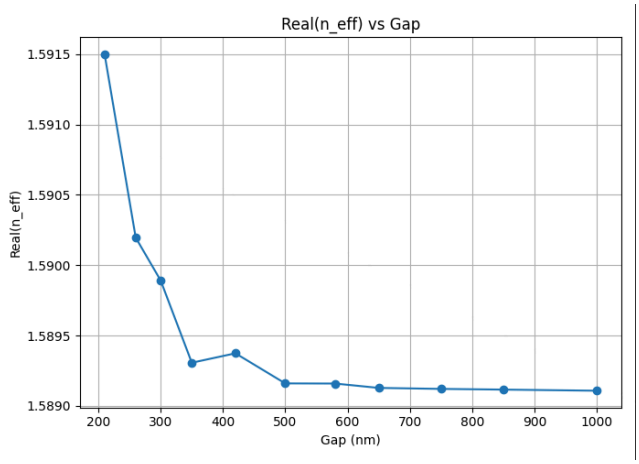
The structure has two stages:

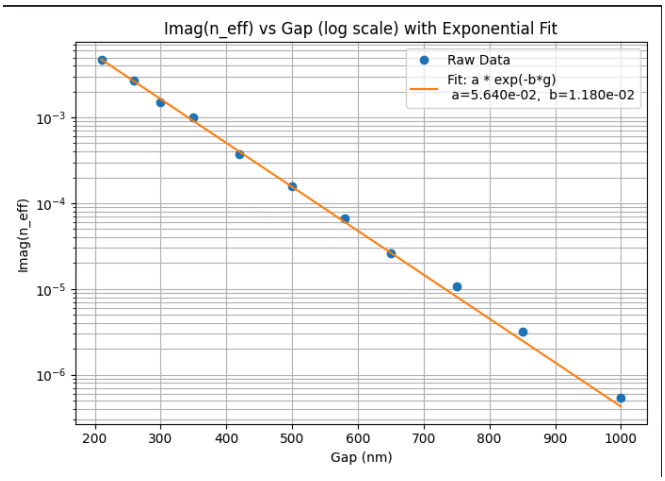
In the first stage, the waveguide mode is evanescently coupled to a Gaussian slab mode. The gap between the waveguide and slab is designed to give a Gaussian mode of a certain waist.
In the second stage, the slab mode is outcoupled to free space Gaussian beam, through an apodized grating structure. The parameters of the grating are optimized through numerical methods to produce Gaussian beam of target beam waist.

b. Design

The material used for waveguide and slab is Silicon Nitride (Si_3N_4 with n = 2.01), and for the cladding is Silicon Dioxide (SiO_2 with n = 1.45). The operating wavelength is 780 nm.

Stage 1: Expander
Using Lumerical's FDE Solver, the effective index of TE_0 is obtained for varying gap lengths. The real and imaginary components are plotted against the gap length. The real part saturates at a certain value with increasing gap, and the imaginary part is exponentially decreasing with gap.

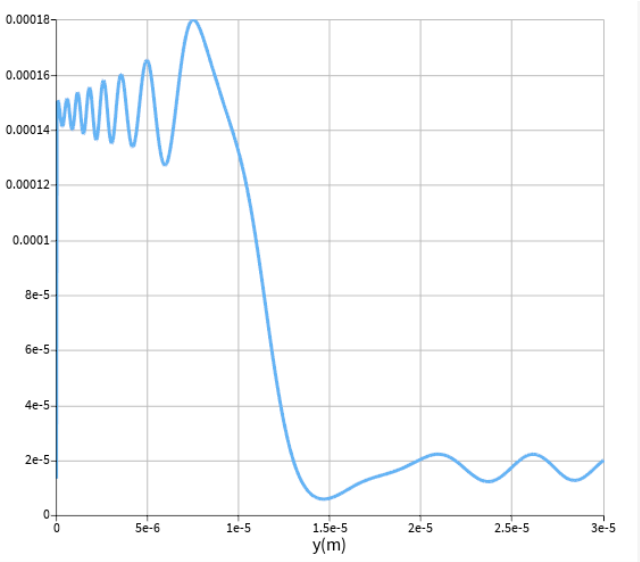




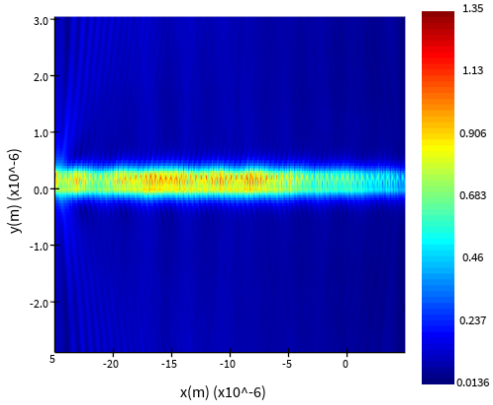
The direction of propagation of slab mode, θ_{tilt} is evaluated from the real part, and the gap profile is given by this formula (with -ve sign and z`^{2} in exp)

$$g(z') = \frac{1}{b}\ln\left\{\frac{1}{a}\frac{\lambda_0}{\sqrt{2}\pi^{3/2}w}\frac{\exp(-2z'/w^2)}{1-\mathrm{erf}\left(\sqrt{2}z'/w\right)}\right\}$$

Here , w_0 = w. sin(θ_{tilt}) is the beam waist of the slab mode.
I imported the data for gap values and simulated the gap structure for 30 μm length to get the following field profile ,along z`.



The reason this came out wrong may be that the gap profile varies very little with length. This means that to accurately capture the field profile, mesh should be made very finite. This requires large compute power to simulate.
However, I also independently tried the apodized grating simulation, with an input Gaussian beam (1D) and grating parameters as given in paper.



This simulation also did not give expected results. There may be two reasons. Firstly, the lateral width of the grating is not specified and using periodic boundary conditions in Lumerical is likely wrong. Second, the beam waist is in 100s of microns, and so, to accurately capture the effect, we need the lateral width to be of that order, which will again, increase the compute power needed for simulation

c. Challenges

Summarizing the above problems, four main challenges are identified.

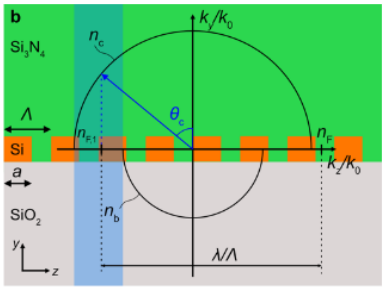
1. The compute power to simulate the structure precisely is huge
2. The structure only controls area, but the efficiency of light coupling is still lost, due to diamond air interface
3. Many design parameters are unclear in the paper
4. The optimisation needed for apodized grating is also unclear and needs to be done, if this method is to be adopted for 532 nm

The complexity of this structure arises in its efforts to generate a perfect Gaussian beam in free space. But we don't exactly need Gaussian beam for our purpose, and hence can look at simpler structures.

4. Design for Ultra Efficient Grating Coupler

a. Working Principle

Alejandro Sánchez-Postigo *et al.* [5] came up with a very novel structure that results in near 100% coupling of light from waveguide to one of the two claddings.

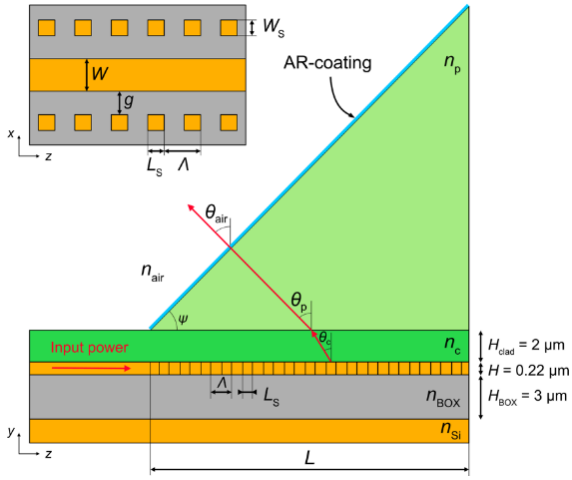


It is derived from coupled mode theory's phase matching condition. Usually, the upper and lower claddings are of the same material, hence the waveguide mode couples symmetrically. But, if we deliberately create an asymmetry in the top and bottom claddings, it is possible to engineer a situation where light from the waveguide is coupled only to one cladding (the one with higher refractive index). This is termed as the 'Single Beam Radiation Condition' and the condition is

$$-n_c < \lambda/\Lambda - n_F < n_b$$

b. Design

In our case, we aim to couple light only into the Diamond. Hence, our upper cladding is fixed. We need to look for a suitable waveguide material and lower cladding. For 532 nm, only two materials have refractive index > diamond's , Gallium Phosphide and Titanium Dioxide.
But, GaP has non zero k (imaginary index) value, leading to very lossy waveguides. So, we proceed with TiO_2. For the lower cladding, a popular choice of SiO_2 is chosen, since it is easy to etch .
The design of the structure is as follows

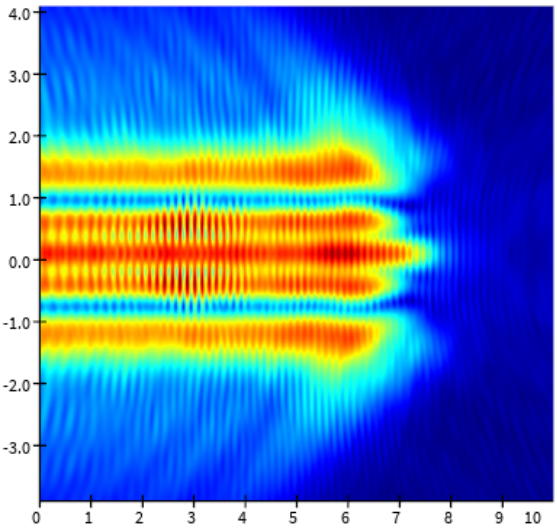


With TiO_2 , the effective index of fundamental mode is 2.439, for W = 200 nm, H = 400 nm.
n_c = 2.425 and n_b = 1.46 . Choosing Λ = 125 nm satisfies the single beam condition , with the beam propagating at 48.5° from vertical. L_s is set to 50 nm arbitrarily, and L to 10 µm for easy simulation.

Now, the task is to adjust the g and W_s values, to get a required grating strength. The grating strength , α_F = -ln(1-P_{rad}/P_{in})/(2.L), and the length needed to emit 99 % of power is 2.3/α_F .
As our diamond sample is a cube of 1mm side, we want 99 % of power to be radiated within 1 mm.

c. Results

An important observation is that lower g and higher w_s gives higher α_F.
Using this and iterating several times, I obtained that for w_s = 400 nm and g = 100 nm, the length of 99% outcoupling is 1.06 mm.
The width of the outcoupled beam is 4 μm, and so roughly 25 such couplers (or less maybe) would be needed to cover the entire sample.



This field profile is inside the diamond. The field profile is cut off at the edge and this may be due to simulation domain effects. In reality, this is not an expected phenomenon.

5. Conclusion and Future scope

This work demonstrated a grating coupler design to couple almost all of the waveguide light into Diamond.
This work can be followed by:

1. Simulating multiple grating couplers in parallel and observing the effect, to determine the number of such couplers needed.
2. Devising a mechanism to efficiently couple the source green light into a TiO_2 waveguide.
3. The design gives freedom to alter many variables. The dependence of field profile, α_F on each of these parameters can be analyzed.

The ANSYS Lumerical project files associated with my work are in this gdrive folder . It also contains the presentation and a Readme file explaining each of the files.

References

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